SACHEM

CLEBURNE, TX

Report No.: 54-11-002
Inspector : Christopher Welch
Employer :
Inspection Date : 02/24/2017

IN-LIEU-OF

INTERNAL

INSPECTION

Pressure Vessel

54-11-002

CLEBURNE

An API Standard 510 Inspection based on client criterion for nondestructive examinations was conducted on vessel 54-11-002 in the SACHEM facility located at CLEBURNE, TX on 02/24/2017. This vessel was originally built to ASME S8 D1. This inspection was conducted in accordance with requirements of the API-510 standard for inspections of Pressure Vessels. The following is a detailed report of the inspection including findings and recommendations.

Inspector Signature
API Certification No. 33958

SACHEM Inc

Cleburne, TX

Report No.: SACHEM METHYL CHLORIDE

Inspector : Timothy Craig

Employer : OilPro Consulting LLC

Inspection Date : 02/11/2016

IN SERVICE

Inspection Report For

54-11-002

Methyl Chloride Tank

An In-Service inspection was completed on the Methyl Chloride Tank 54-11-002 in the SACHEM Inc facility located at Cleburne, TX on 02/11/2017. The inspection was conducted in accordance with client criterion for NDE which included visual & ultrasonic examinations. The tank was built to API 510 specification for Pressure Vessels. This inspection was conducted in accordance with requirements of the API-653 standards.

The tank bottom was inspected from 135 degrees to 225 degrees with UT and VT technology in order to assess the underside condition of the tank bottom. No underside indications were found reflecting significant underside corrosion issues. No pits were found during the Visual Inspection. The tank shell wall thickness ranged from 0.814 to 0.836 with the exception of 3 areas where the tank coating had been removed and only a thin primary coat was present. In these locations, the shell wall thickness ranged from 0.802 to 0.806. Within 6 inches of the tank supports struts, the shell wall thickness increase to a range from 0.850 to 0.860.

INLIEU OF REFERENCED FROM API 510

II. Internal/On-Stream Inspection (includes UT measurements): (1/2 Life w/ Max of 10 Years)
(API 510 section 5.5.2, 5.5.5, 5.6.2 & 6.5)

A. Vessel internal inspection should be scheduled every 10 years or 1/2 corrosion rate life. Where the estimated remaining life is <4 years, the next inspection interval may be scheduled at the full life expected up to 2 years max.

B. The 10 years shall be the 10 years of actual service for vessels in noncontiguous service isolated from the process (isolated so that they are not exposed to corrosive environments (i.e., inert gases with noncorrosive hydrocarbons)).

C. Except as noted below, internal inspection is normally the preferred method of inspection and shall be conducted on vessels subject to significant localized corrosion and other types of damage. When the authorized inspector believes it is warranted an on-stream inspection may be substituted for internal inspection in the following situations:

✔ 1. When do to vessel size, configuration, or lack of access makes vessel entry for inspection physically impossible.

✔ 2. When the general corrosion rate is less than 0.005 inch per year and remaining life is >10 years, and all of the following conditions are met:

✔ a. The corrosive characteristics of the vessel contents, including the effect of trace components, has been established by at least 5 years of the same or comparable service experience with the type of same contents.
✔ b. No questionable condition from the external inspection is found.
✔ c. The operating temperature of the steel shell does not exceed the lower temperature limits for the creep-rupture range of the material of construction.
✔ d. The vessel is not subject to environmental cracking or hydrogen damage from the vessel contents.
✔ e. The vessel walls are not strip-lined or plate-lined.

✔ D. A representative number of thickness measurements must be conducted on each vessel components such as shells, heads and nozzles to satisfy the requirements for an internal (or on-stream inspection). The remaining life of the vessel for establishing inspection intervals is to be based on the most limited component.

All of the above have been satisfied to achieve an in lieau of under the following conditions :

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1.0 EXECUTIVE SUMMARY

An API Standard 510 inspection of pressure vessel 54-11-002 located in CLEBURNE, TX, was conducted on 02/24/2017. This inspection was made to collect data in order to evaluate the mechanical integrity and fitness for service of the vessel. No major problems were noted. The vessel is in satisfactory mechanical condition for continued service.Minor discrepancies are listed in Section 3.0 Inspection Results and 4.0 Recommendations.

TABLE A

Component	*Nominal Design Thickness (in.)*	*Actual Measured Thickness (in.)*	*Minimum Required Thickness (in.)*	*Design MAWP (psi) Internal*	*Calculated MAWP (psi) Internal*	*Remaining Life (years)*
Shell 1	0.822	0.780	0.751	225	227.2	27.2
North Head	0.530	0.516	0.433	225	264.8	236.3
South Head	0.530	0.519	0.433	225	267.2	311.7
				225		

Next Inspection:

Next external inspection is due by: 2/24/2022
Next internal inspection is due by: 2/24/2027
Next UT inspection is due by: 2/24/2027

2.0 VESSEL DATA

Vessel Data:		***General Data:***	
MAWP (psi):	225	***Product:***	METHYLCLORIDE
Design Temp. (degF):		***Year Built:***	1977
MDMT (degF):	-20	***NB No.:***	51113
Oper. Press.(psi):	125	***Const Code***	ASME S8 D1
Oper. Temp. (degF):		***Vessel Config.:***	Horizontal
Material Type:	Carbon Steel	***Head Type:***	Torispherical
Inside Dia. (inch):	130.250	***Insul. Type:***	None
Length (inch)	561.875	***Insul. Thk (inch):***	

MAWP (psi):	***Oper. Press.(psi):***
Design. Temp. (degF):	***Oper. Temp.(degF):***
Material:	***Medium:***

APPENDIX A

Thickness Measurement Record

1) Component Thickness Measurements

API-510 PRESSURE VESSEL COMPONENT THICKNESS RECORD

Inspection Data

Report No	Client	Inspector	Vessel	Date
54-11-002	SACHEM	Christopher Welch	54-11-002	02/24/2017

Components with Vert. Axis: tml-1 N., tml-2 E., tml-3 S., tml-4 W. (Drawing N.)

Components with Horz. Axis: tml-1 Top, tml-2 Side, tml-3 Bttm., tml-4 Side (Clock Wise)

CML	*Comp ID*	*Location*	*Service*	*tml-1*	*tml-2*	*tml-3*	*tml-4*	*t act*
001	Bttm Head			0.516				0.516
002	Bttm Head			0.526				0.526
003	Bttm Head			0.528				0.528
004	Vessel Shell			0.529				0.529
005	Bttm Head			0.526				0.526
006	Vessel Shell			0.795	0.793	0.792	0.793	0.792
007	Vessel Shell			0.810	0.812	0.817	0.819	0.810
008	Vessel Shell			0.815	0.818	0.822	0.820	0.815
009	Vessel Shell			0.815	0.815	0.815	0.818	0.815
010	Vessel Shell			0.814	0.812	0.810	0.809	0.809
011	Vessel Shell			0.797	0.788	0.787	0.782	0.782
012	Vessel Shell			0.790	0.786	0.788	0.784	0.784
013	Vessel Shell			0.781	0.788	0.786	0.780	0.780
014	Vessel Shell			0.797	0.803	0.805	0.806	0.797
015	Vessel Shell			0.804	0.812	0.810	0.811	0.804
016	Vessel Shell			0.802	0.811	0.810	0.812	0.802
017	Vessel Shell			0.808	0.811	0.813	0.811	0.808
018	Vessel Shell			0.793	0.793	0.790	0.791	0.790
019	Vessel Shell			0.791	0.792	0.795	0.792	0.791
020	Vessel Shell			0.793	0.795	0.796	0.794	0.793
021	Vessel Shell							
022	Vessel Shell			0.784	0.788	0.787	0.788	0.784
023	Vessel Shell							
024	Vessel Shell							
025	Vessel Shell							
026	Top Head			0.519				0.519
027	Top Head			0.521				0.521
028	Top Head			0.525				0.525
029	Top Head			0.523				0.523
030	Top Head			0.526				0.526

APPENDIX B

Mechanical Integrity Calculation

1) Report Cylindrical Shell Calculations
2) Report Formed Head Calculations
3) Report Nozzle Calculation

API-510 PRESSURE VESSEL SHELL EVALUATION

MINIMUM THICKNESS, REMAINING LIFE, PRESSURE CALCULATIONS

Report No.	Client	Inspector	Vessel	Date
54-11-002	SACHEM	Christopher Welch	54-11-002	2/24/2017

Shell 1			Material						Material		
			CS - A612-B						CS - A612-B		
Temp.	MAWP	SH	SG	D	t nom	Temp.	MAWP	SH	SG	D	t nom
	225	0.0	0.73	130.250	0.822			5.0	0.73	130.250	

Minimum Thickness Calculations

Shell 1		Internal	PR/(SE-0.6P) = t				Internal	PR/(SE-0.6P) = t	
P	R	S	E	t	P	R	S	E	t
225	65.125	23100	0.85	0.751		65.125	20200	0.85	

Remaining Life Calculations

Shell 1	t prev	t act	t min	y		t prev	t act	t min	y
	0.822	0.780	0.751	40.0					

Ca = t act - t min	=	0.029 (inch)	Ca = t act - t min	=	(inch)
Cr = t prev - t act / Y	=	0.00105 (in/year)	Cr = t prev - t act / Y	=	(in/year)
RL= Ca / Cr	=	27.2 (year)	RL= Ca / Cr	=	(year)

Next Inspection (Yn) = 10 (years)

MAWP Calculations

Shell 1	MAP - Next Inspection		MAP - Next Inspection	
Where t = t act - 2YnCr =	0.759	(inch)	Where t = t act - 2YnCr =	(inch)
SEt/R+0.6t = P =	227.2	(psi)	SEt/(R+0.6t) = P =	(psi)
P-(SH*.433*SG) = MAWP =	227.2	(psi)	P-(SH*.433*SG) = MAWP =	(psi)

API-510 PRESSURE VESSEL SHELL EVALUATION				
MINIMUM THICKNESS, REMAINING LIFE, PRESSURE CALCULATIONS				
Report No.	***Client***	***Inspector***	***Vessel***	***Date***
54-11-002	SACHEM	Christopher Welch	54-11-002	2/24/2017

Variable Definitions for Shell Calculations:

A = factor determined from Fig. G in Subpart 3 of Section II, Part D and used to enter the applicable material chart in Subpart 3 of Section II, Part D. For the case of cylinders having Do /t values less than 10, see UG-28(c)(2).

B = factor determined from the applicable material chart in Subpart 3 of Section II, Part D for maximum design metal temperature, psi [see UG-20(c)]

Ca = remaining corrosion allowance of the vessel part under consideration, in inches.

Cr = corrosion rate of the vessel part under consideration, in inches per year.

D = inside diameter of the shell course under consideration, in inches.

E = (Internal Calculations) lowest efficiency of any joint in the shell course under consideration. For welded vessels, use the efficiency specified in UW-12.

E = (External Calculations) - Modulus of Elasticity (MOE) at operating temperature for specified material,
P = the design maximum allowable internal working pressure, including static head pressure, in psi.

Pa = maximum allowable external working pressure (includes jacket pressure and vessel internal negative pressure) in psi.

R = inside radius of the shell under consideration, in inches.

Ro = outside radius of the shell under consideration, in inches.

RL = estimated remaining life of the vessel part under consideration, in years.

S = maximum allowable stress value, in psi.

SH = static head, in feet

SG = specific gravity of vessel product.
t = thickness of the vessel part under consideration, variable related to applicable calculation used therein, in inches.

t act = actual thickness measurement of the vessel part under consideration, as recorded at the time of inspection, in inches.

t min = minimum required thickness of vessel part under consideration, as the nominal thickness minus

the design corrosion allowance or the calculated minimum required thickness at the design MAWP at the coinciding working temperature, in inches.

t nom = design nominal thickness of head, in inches.
t prev = previous thickness measurement of the vessel part under consideration, as recorded at last inspection or nominal thickness if no previous thickness measurements, in inches.

Y = time span between thickness readings or age of the vessel if t nom is used for t prev, in years

Yn = estimated time span to next inspection of the vessel part under consideration, in years.

API-510 PRESSURE VESSEL HEAD EVALUATION

MINIMUM THICKNESS, REMAINING LIFE, PRESSURE CALCULATIONS

Report No.	Client	Inspector	Vessel	Date
54-11-002	SACHEM	Christopher Welch	54-11-002	2/24/2017

Vessel Head(s)

North Head and South Head

MAWP	D	T	E	SG1	SG2
225	130.250		0.85	0.73	0.73

MAWP	D	T	SH	SG3	E
225	130.250				0.85

Head ID	Head Type	t nom
North Head	Hemispherical	0.530

Material	S	SH	P
CS - A612-B	20200	10.0	228.2

Head ID	Head Type	t nom
South Head	Hemispherical	0.530

Material	S	SH	P
CS - A612-B	20200	10.0	228.2

Head ID	Head Type	t nom

Material	S	P
		225.0

For Torispherical Heads

L = 65.137 R = 65.137 M = 1	L = 65.137 R = 65.137 M = 1	L = R = M =

Minimum Thickness Calculations

Internal;	Hemispherical Head	2:1 Ellipsodial Head	Torispherical Head

PL/(2SE-0.2P) = t min PD/(2SE-0.2P) = t min (knl) PLM/(2SE-0.2P) = t min (knl)

P0.9D/(2S-0.2P) = t min (crwn) PL/(2S-0.2P) = t min (crwn)

(reference supplemental calcs for other head type formulas)

North Head	Hemispherical	t min = 0.433 (inch)
South Head	Hemispherical	t min = 0.433 (inch)
		t min = (inch)

API-510 PRESSURE VESSEL HEAD EVALUATION

MINIMUM THICKNESS, REMAINING LIFE, PRESSURE CALCULATIONS

Report No.	Client	Inspector	Vessel	Date
54-11-002	SACHEM	Christopher Welch	54-11-002	2/24/2017

Remaining Life Calculations

North Head	*t prev*	*t act*	*t min*	*y*
	0.530	0.516	0.433	40.0

Ca = t act - t min	=	0.083	(inch)
Cr = t prev - t act / Y	=	0.00035	(in/year)
RL= Ca / Cr	=	236.3	(years)

	t prev	*t act*	*t min*	*y*

Ca = t act - t min	=		(inch)
Cr = t prev - t act / Y	=		(in/year)
RL= Ca / Cr	=		(years)

South Head	*t prev*	*t act*	*t min*	*y*
	0.530	0.519	0.433	40.0

Ca = t act - t min	=	0.086	(inch)
Cr = t prev - t act / Y	=	0.00028	(in/year)
RL= Ca / Cr	=	311.7	(years)

Next Inspection (Yn) 10 (year)

MAWP Calculations *(reference supplemental calcs for other head type formulas)*

MAP	*Hemispherical Head*	*2:1 Ellipsodial Head*	*Torispherical Head*
Internal	*2SEt/(R+0.2t) = P*	*2SEt/(D+0.2t) = P (knl)*	*2SEt/(LM+0.2t) = P (knl)*
		2St/(0.9D+0.2t) = P (crwn)	*2St/(L+0.2t) = P (crwn)*

Where t = t act - 2YnCr ***Where P = MAP at the Next Inspection***

Where MAWP = P-(SH*.433*SG)

North Head	*Hemispherical*	*t =*	0.509	*(inch)*	*P=* 268.0	*(psi)*	*MAWP =* 264.8	*(psi)*
South Head	*Hemispherical*	*t =*	0.514	*(inch)*	*P=* 270.3	*(psi)*	*MAWP =* 267.2	*(psi)*
		t =		*(inch)*	*P=*	*(psi)*	*MAWP =*	*(psi)*

API-510 PRESSURE VESSEL SHELL EVALUATION				
MINIMUM THICKNESS, REMAINING LIFE, PRESSURE CALCULATIONS				
Report No.	Client	Inspector	Vessel	Date
54-11-002	SACHEM	Christopher Welch	54-11-002	2/24/2017

Variable Definitions for Shell Calculations:

A = factor determined from Fig. G in Subpart 3 of Section II, Part D and used to enter the applicable material

chart in Subpart 3 of Section II, Part D. For the case of cylinders having Do /t values less than 10, see UG-28(c)(2).

B = factor determined from the applicable material chart in Subpart 3 of Section II, Part D for maximum design metal

temperature, psi [see UG-20(c)]
Ca = remaining corrosion allowance of the vessel part under consideration, in inches.

Cr = corrosion rate of the vessel part under consideration, in inches per year.

D = inside diameter for the head skirt, in inches.

Do = outside diameter for the head skirt, in inches.

E = (Internal Calculations) lowest efficiency of any joint in the vessel part under consideration. For welded vessels, use the efficiency specified in UW-12.

E = (External Calculations) - Modulus of Elasticity (MOE) at operating temperature for specified material.

h = inside height for the head, in inches.

ho = outside height for the head, in inches.

H = inside head height from tangent, in inches.

K = factor depending on head proportions D/2h and determined as 1/6[2+(D/2h)^2]

Ko = factor depending on head proportions Do/2ho (see table UG 33.1)
L = Hemi. Hds - inside spherical or crown radius of the head under consideration, in inches,. Ellip. and Tor.

Hds - inside spherical or crown radius, in inches,

M = a factor used in the formulas for torispherical heads where M= 0.25(3+(L/r)^.5)

P = the design maximum allowable internal working pressure, including static head pressure, in psi.

Pa = maximum allowable external working pressure, including static head pressure, in psi.

r = inside knuckle radius, in inches.

R = inside radius of the shell under consideration, in inches.

Ro = Hemi - outside radius in inches. Ellip. - the equivalent outside spherical radius taken as KoDo.

Tor. - outside radius of the crown portion of the head, in inches.

API-510 PRESSURE VESSEL SHELL EVALUATION				
MINIMUM THICKNESS, REMAINING LIFE, PRESSURE CALCULATIONS				
Report No.	**Client**	**Inspector**	**Vessel**	**Date**
54-11-002	**SACHEM**	**Christopher Welch**	**54-11-002**	**2/24/2017**

RL = estimated remaining life of the vessel part under consideration, in years.

S = maximum allowable stress value, in psi.

SH = static head, in feet

SG = specific gravity of vessel product.

t = thickness of the vessel part under consideration, variable related to applicable calculation used therein, in inches.

t act = actual thickness measurement of the vessel part under consideration, as recorded at the time of inspection, in inches.

t min = minimum required thickness of vessel part under consideration, as the nominal thickness minus the design
corrosion allowance or the calculated minimum required thickness at the design MAWP at the coinciding working
temperature, in inches.

t nom = design nominal thickness of head, in inches.

t prev = previous thickness measurement of the vessel part under consideration, as recorded at last inspection or
nominal thickness if no previous thickness measurements, in inches.

Y = time span between thickness readings or age of the vessel if t nom is used for t prev, in years.

Yn = estimated time span to next inspection of the vessel part under consideration, in years.

API-510 PRESSURE VESSEL NOZZLE EVALUATION

MINIMUM THICKNESS, REMAINING LIFE, PRESSURE CALCULATIONS

Report No	Client	Inspector	Vessel	Date
54-11-002	SACHEM	Christopher Welch	54-11-002	02/24/2017

Minimum Thickness Determinations:

a) The following nozzle minimum thicknesses are based on current ASME Standards. Minimum thickness allowed for nozzles walls are based on standard pipe thicknesses minus 12.5% or connecting shell/head required thickness whichever is smaller. (ASME Sect VIII, UG-45)

Size (inch)	3/4	1	1.5	2	3	4	6	8	10	12	>12
tmin (inch)	0.100	0.116	0.127	0.135	0.189	0.207	0.245	0.282	0.319	0.328	0.328

b) Large size nozzles or nozzles subject to high pressures are calulated per ASME Sect VIII, DIV 1, UG-27 as follows: PR/(SE-0.6P) = t.

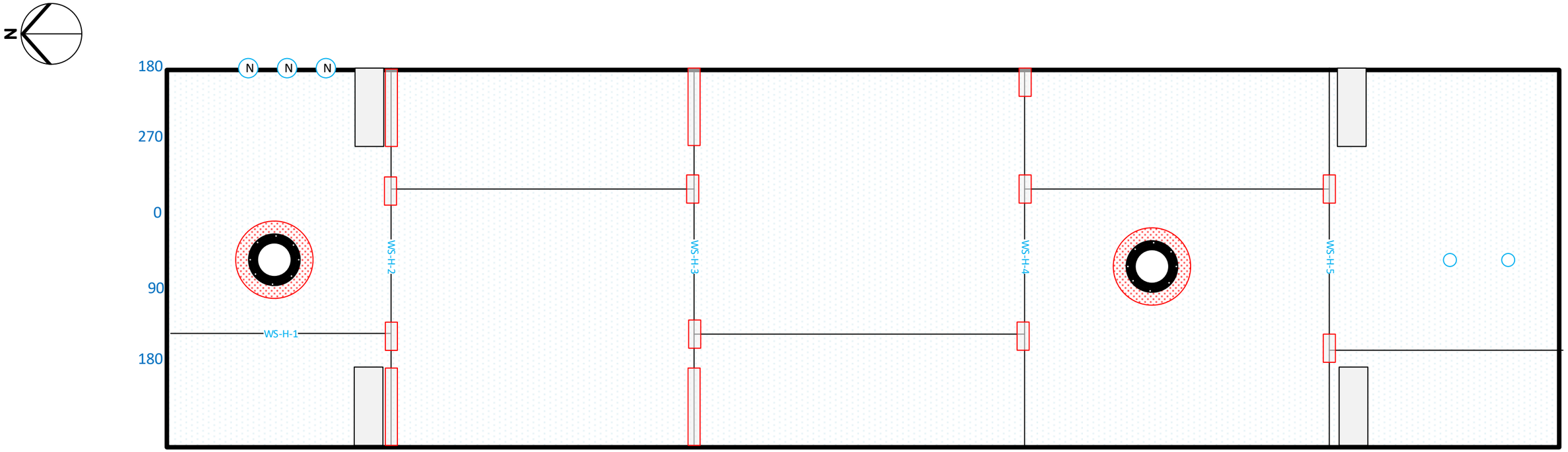
Nozzle Remaining Life Calculations:

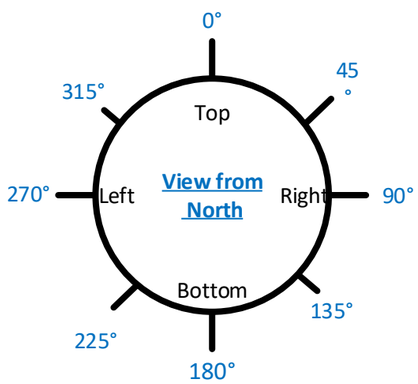
CML	Noz ID.	Size (inch)	Material	Age (year)	t prev (inch)	t act (inch)	t min (inch)	Ca (inch)	Cr (inch/Yr)	RL (year)

APPENDIX C

Inspection Drawings

1) Vessel Layout Drawing





UT SHEARWAVE

Note:
0°- Top of Vessel
Reading should be taken in clockwise fashion starting at East end

Weld Seam	Point No.	Original Thickness	Minimum Allowed Thickness	Date	Date2	Date3	Date4	Date5	Date6	Date7	Date8
	1										
	2										
	3										
	4										
	5										
	6										
	7										
	8										
	9										
	10										

OilPro CONSULTING
A LIMITED LIABILITY COMPANY

Instrument Serial No.			Inspection Date	
Transducer Serial No.			Inspection Type	Thickness Readings
Calibration Block			Company	
Flat or Curved		Serial No.	Technician	
Material		Manufacturer		
Type				

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Scale: None

APPENDIX D

Inspection Checklist

1) API 510 Checklist

API-510 PRESSURE VESSEL INSPECTION CHECKLIST				Inspector:	Christopher Welch
Company:	SACHEM			Cert No.:	33958
Vessel:	54-11-002	Report No.:	54-11-002	Date:	2/24/2017

1 FOUNDATION

1.1 Steel Members

a.		Visually inspect for pitting, and corrosion.
b.		Check fnd bolts secure with minimum thrd engagement
c.		Check for coating failures
d.		Check attachment welds for cracking and corrosion.

1.2 Containment

a.		Inspect the area for buildup of trash, vegetation and obstructions.
b.		Inspect sump drain operation.
c.		Check that runoff rainwater drains away from the tank.
d.		Describe type of construction - Earthen, Concrete, Asphalt, Gravel
e.		Inspect condition of containment.

1.3 Concrete Foundation Supports

a.		Inspect for broken concrete, spalling and cracks.
b.		Inspect for erosion under foundation.
c.		Check for settlement around perimeter of tank.

1.4 Wooden Saddle Support

a.		Check for degradaded members (split, broken, dry rotted et.).
b.		Inspect for errosion and vegetation tank fnd.
c.		Check for settlement around perimeter of tank.

2 SHELLS

2.1 External Visual Inspection

a.		Visually inspect shell surface for paint failures, pitting, corrosion, denting, out-of-round and part deformation.
b.		Clean angles and other components that form catch basins and check for degradation and corrosion..
c.		Inspect nozzle penetration surfaces and welds for corrosion, cracking and deformation.
d.		Check for broken, unused insulation rod supports causing corrosion nodes.
e.		Visually inspect weld joints for cracking, pitting, corrosion and signs of leaking (product residue).
f.		Perform dye penetrant or magnetic particle tests if leaks or cracks are suspected.
g.		Check for proper grounding

2.2 Internal Visual Inspection

a.		Check atmospheric conditions, fill out and post safe entry permit form.
b.		Appropriate and wear required PPE for safe entry.
c.		Inspect shell surfaces for coating failures, pitting and corrosion.
d.		Inspect baffle plate surfaces &weld attachments for cracking, pitting, or corrosion. Check bolts are secure with minimum thread engagement.
e.		Inspect heating coils surfaces and weld attachments, for cracking, pitting, corrosion. Check bolts are secure with minimum thread engagement.
f.		Inspect pressure containing weld joints for cracking, pitting, and corrosion.
g.		Perform dye penetrant or magnetic particle tests if cracks are suspected.

3 SHELL APPURTENANCES

3.1 Manways and Nozzles

a.		Inspect for cracks or signs of leakage on weld joints at nozzles, manways, and reinforcing plates.
b.		Inspect for shell plate dimpling around nozzles, caused by excessive pipe deflection.
c.		Inspect for flange leaks and leaks around bolting.
d.		Check flange bolts are secure and have minimum thread engagement.
e.		Inspect sealing of insulation around manways and nozzles.
f.		Check for inadequate manway flange and cover thickness on mixer manways.
g.		Check exposed flange and cover faces.

3.2 Relief Devices

a.		Inspect for flange leaks and leaks around bolting.				
b.		Check flange bolts are secure and have minimum thread engagement.				
c.		Record inlet and outlet sizes sizes.	Inlet		Outlet	
d.		Check that relief system outlet discharges to safe location (outside of building).				
e.		Record certification date				
f.		Record pressure setting				
g.		Record type and ID				

3.3 Shell-Mounted Sample Station

a.		Inspect sample lines for function of valves and plugging of lines, including drain or return-to-tank line.
b.		Check circulation pump for leaks and operating problems.
c.		Test bracing and supports of sample system and equipment.

API-510 PRESSURE VESSEL INSPECTION CHECKLIST				Inspector:	Christopher Welch
Company:	SACHEM			Cert No.:	33958
Vessel:	54-11-002	Report No.:	54-11-002	Date:	2/24/2017

3.4 Heater (Steam Coils)

a.		Inspect condensate drain for presence of oil, indicating leakage.

3.5 Mixer/Agitator

a.		Inspect for proper mounting flange and support.
b.		Inspect for leakage.
c.		Inspect condition of power lines and connections to mixer.

3.6 Deck Plate and Grating

a.		Inspect deck plate for corrosion-caused thinning or holes (not drain holes) and paint failure.
b.		Inspect plate-to-frame weld for rust scale buildup.
c.		Inspect grating for corrosion-caused thinning of bars and failure of welds.
d.		Check grating tie down clips. Where grating has been retrofitted to replace plate, measure the rise of the step below and above the grating surface and with other risers on the stairway.

3.7 Stairway Stringers/Rungs/Treads

a.		Inspect stairway stringers, rungs and treads for corrosion, paint failure and weld failure.
b.		Inspect stairway supports to shell welds and reinforcing pads.
c.		Inspect steel support attachment to concrete base for corrosion.

4 VESSEL JACKET

4.1 Jacket Shell

a.		Identify type of jacket (half pipe, cylindrical, dimpled and spot welded, etc.)
b.		Measure and record pitch distances.
c.		Visually inspect shell surface for paint failures, pitting, corrosion, denting, out-of-round and part deformation.
d.		Clean angle support rings and inspect for corrosion and thinning on plate, annular space and welds.
e.		Inspect the shell-to-foundation seal or barrier.
f.		Check for broken, unused insulation rod supports causing corrosion nodes.
g.		Visually inspect for weld joints for cracking, pitting, corrosion and signs of leaking (product residue).
h.		Perform dye penetrant or magnetic particle test if leaks or cracks are suspected.

5 INSULATION

5.1 Visual Inspection

a.		Check for holes, missing portions, deterioration due to corrosion or abuse.
b.		Check for sufficient sealing, especially around vessel appurtenances.
c.		Check for wetness

6 PRESSURE /TEMPERATURE INDICATORS

6.1 Physical and Operating Conditions

a.		Check that indicators are securely attached and operating properly.
b.		Check that indicators are in accessible locations and readable.
c.		Check that indicators have current calibration date
d.		Check for any damage, and corrosion build up.

7 ASME/NAME PLATE DATA

7.1 Physical Condition

a.		Check that ASME plate is securely attached
b.		Check that ASME plate is in accessible location and readable.

API-510 PRESSURE VESSEL INSPECTION CHECKLIST				Inspector:	Christopher Welch
Company:	SACHEM			Cert No.:	33958
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7.2 Record Following ASME Data

a.		Board Number	
b.		Serial Number	
c.		Radiography Examination	
d.		MAWP	
e.		MDMT	
f.		Nominal Shell Thickness	
g.		Nominal Head Thickness	
h.		Nominal Hieght	
i.		Head Material	
j.		Shell Material	
k.		Jacket Material	
l.		Manufacturer	
m.		Year Built	

8 VESSEL LAYOUT DRAWINGS

8.1 Record Following Data

a.		Head and skirt lengths and type.
b.		Shell lines of support (Head Tang, Braces, Jckt Closures)
c.		Foundation support member dimensions and orientation
d.		Inside/Outside diameter.
e.		Vessel part nominal thicknesses.
f.		Nozzle layouts, sizes and uses.

9 UT THICKNESS READINGS

9.1 Measure and Record Vessel Part Thicknesses

a.		Heads - (4) locations, (1) in each quadrant.
b.		Shell - (4) locations, (1) in each quadrant for each course.
c.		Jacket - As required (at least (4) locations).
d.		Nozzles - (4) locations, (1) in each quadrant for 2

Notes:

APPENDIX E

Inspection Photographs

1) Photo Files

APPENDIX F

Manufacturers Data Sheets

1) U1 Mfg Data Sheet
2) Engineering Design Drawings

APPENDIX G

NDE Records

1) UT Calibration Record
2) NDE Technician Certification
3) API Inspector Certification